

Question Number	Answer	Additional Guidance	Mark
3(a)	An answer that includes the following points: • {the tendency / ability / potential of water (molecules) to move (out of a solution / cell) / described} (1) • {water moves from higher to lower WP / more free water, greater its value / pure water zero / always negative} (1)	concentration of free water molecules ACCEPT free water molecules are ones not associated with solute	(2)

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3(b)(i)	An answer that includes the following points: • Valid; by removing excess {water/ solution} (before weighing) / use of electronic balance (1) • Accurate; blotting dry for same time / by suitable use of balance described (1)	not dried unless clear just the surface e.g., zeroed, calibrated, balance to suitable number of decimal places	(2)

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3(b)(ii)	sucrose (solution) concentration	ACCEPT tissue type	(1)

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3(b)(iii)	An explanation that includes the following points: {water would {enter / leave} the potato more quickly than the yam / potato mass would {increase / decrease} more than the yam (in same time)} (1) {because increased temperature makes (water) molecules move around more / (water) molecules have more kinetic energy} (1) - this would make (method / results) invalid (1)	ACCEPT quicker result / water enters easily / osmosis happens quicker Ignore faster reaction	(3)

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3(c)(i)	A graph with the following features: - x and y-axis labelled (1) - suitable scale on y axis (1) - line of best fit for yam (1)	percentage change in mass sucrose concentration / M dm⁻³ Yam Potato $y = -40.443x + 28.438$ $y = -43.926x + 13.723$	(3)

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3(c)(ii)	A calculation including the following steps. • correct gradient calculated (1) • correct reading of intercept (1) • correct equation (1)	e.g., -45 (accept -43 to -45.4) e.g., 13 (accept 12-14) e.g., $y = -45x + 13$ allow ecf equation	(3)

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3(c)(iii)	• 0.30 (1) • -800 (kPa) (1)	Allow 0.28, 0.29 ecf for misread of first graph	(2)

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3(c)(iv)	An answer that includes the following points: • solute {concentrations / potentials} free water concentration} different in cells (1)	ACCEPT named relevant solute	(1)